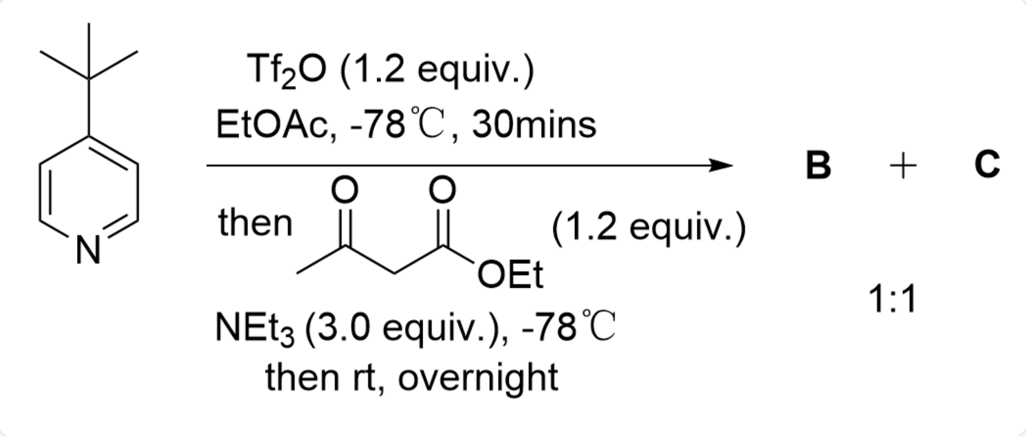
Question

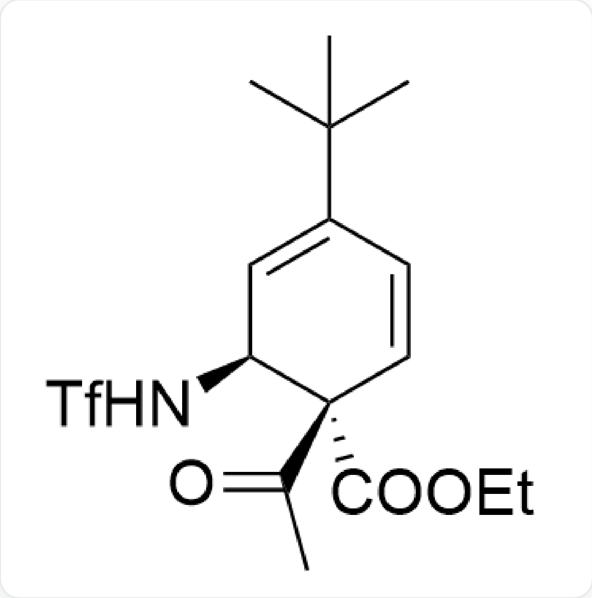


CC(C)(C)C1=CC=NC=C1>$\mathrm{Tf_2O}$(1.2 equiv),EtOAc, reaction at $-78^{\circ}\mathrm{C}$ for 30 mins; then CC(CC(OCC)=O)=O(1.2 equiv),$\mathrm{NEt_3}$(3.0 equiv), at $-78^{\circ}\mathrm{C}$; then rt, overnight>[$\mathbf{B}$].[$\mathbf{C}$], the ratio of products $\mathbf{B}$ and $\mathbf{C}$ is approximately 1:1

Given that the reaction yields products $\mathbf{B}$ and $\mathbf{C}$ in an approximate ratio of 1:1 and their molecular formulas are different, and the molecular formula of product $\mathbf{C}$ is $\mathrm{C_{16}H_{22}F_3NO_5S}$, without considering enantiomers, provide the structural formula of product $\mathbf{C}$.

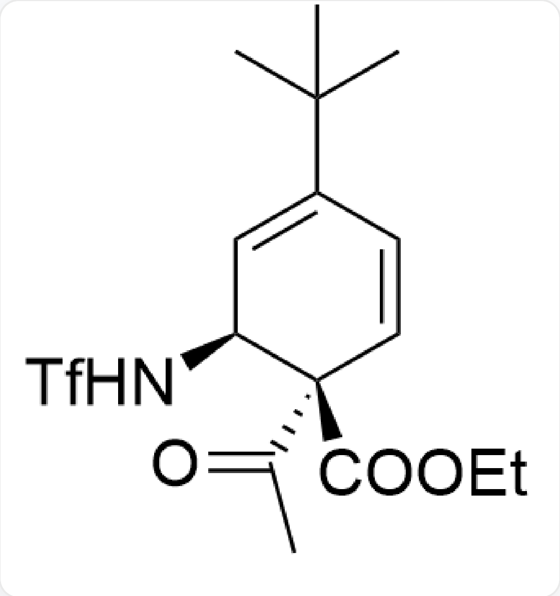
A. All other options are incorrect

B.



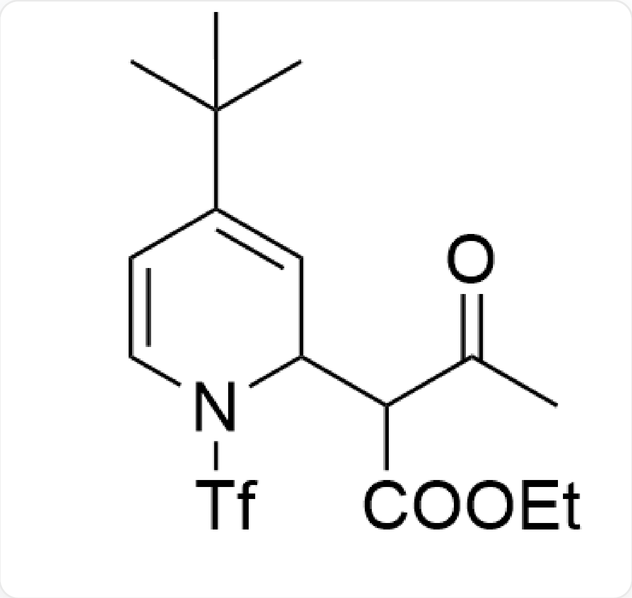
CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@@](C(C)=O)(C(OCC)=O)C=C1

C.



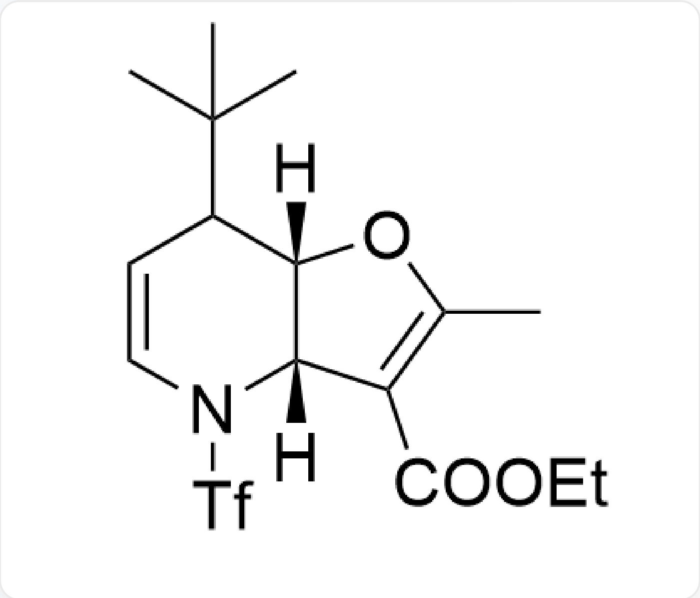
CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@](C(C)=O)(C(OCC)=O)C=C1

D.



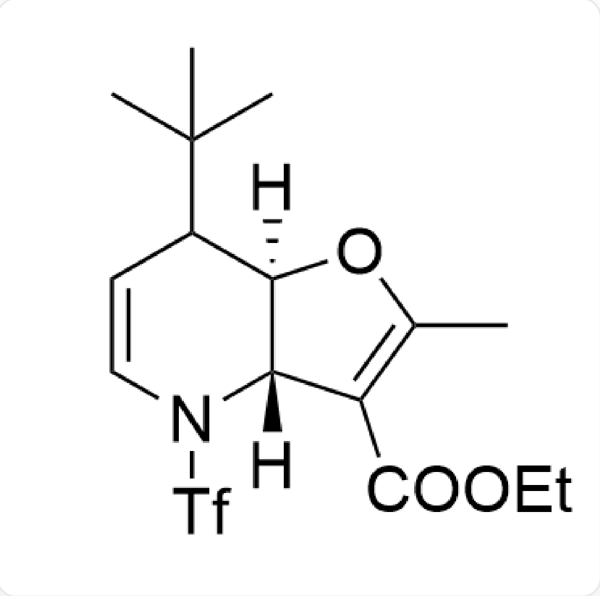
CC(C)(C)C1=CC(C(C(OCC)=O)C(C)=O)N(S(=O)(C(F)(F)F)=O)C=C1

E.



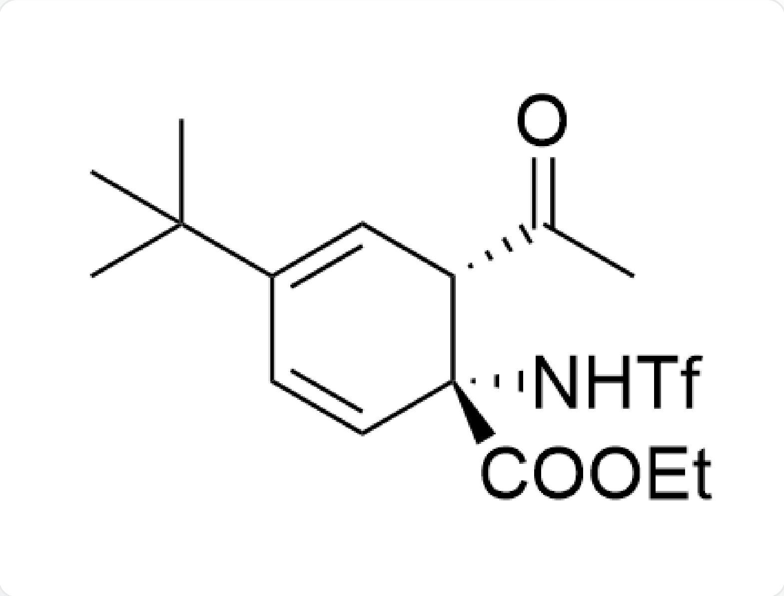
CC(C)(C)C1[C@@]2([H])[C@@](C(C(OCC)=O)=C(C)O2)([H])N(S(=O)(C(F)(F)F)=O)C=C1

F.



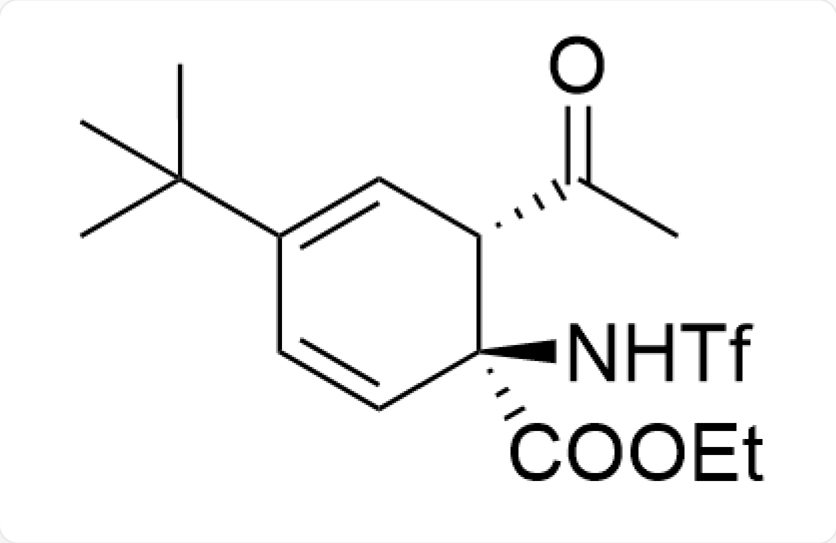
CC(C)(C)C1[C@]2([H])[C@@](C(C(OCC)=O)=C(C)O2)([H])N(S(=O)(C(F)(F)F)=O)C=C1

G.



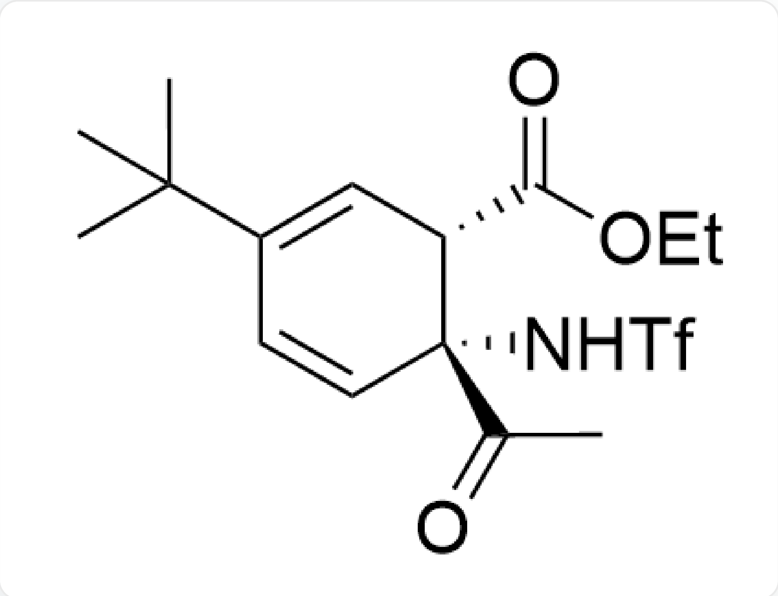
CC(C)(C)C1=C[C@H](C(C)=O)[C@](C(OCC)=O)(NS(=O)(C(F)(F)F)=O)C=C1

H.



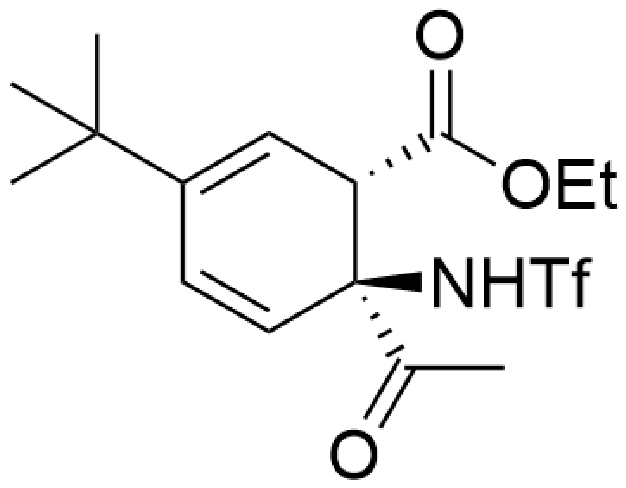
CC(C)(C)C1=C[C@H](C(C)=O)[C@@](C(OCC)=O)(NS(=O)(C(F)(F)F)=O)C=C1

I.



CC(C)(C)C1=C[C@H](C(OCC)=O)[C@](C(C)=O)(NS(=O)(C(F)(F)F)=O)C=C1

J.



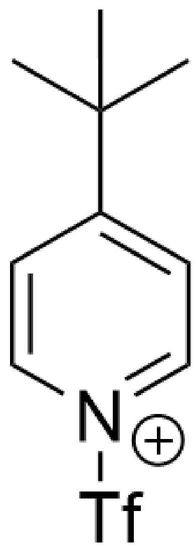
CC(C)(C)C1=C[C@H](C(OCC)=O)[C@@](C(C)=O)(NS(=O)(C(F)(F)F)=O)C=C1

Answer

Correct Answer: C

Detailed Explanation

First, observe the first reaction. It is obvious that the substrate reacts with Tf_2O to obtain intermediate $\mathbf{1}$

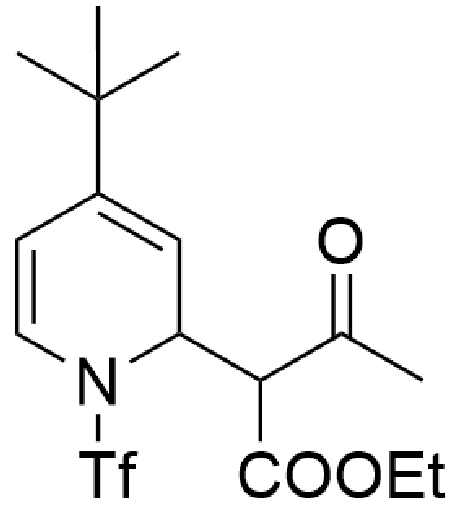


Intermediate $\mathbf{1}$: CC(C)(C)C1=CC=[N+](S(=O)(C(F)(F)F)=O)C=C1

CHECKPOINT 1 PTS

Intermediate $\mathbf{1}$: CC(C)(C)C1=CC=[N+](S(=O)(C(F)(F)F)=O)C=C1

Then, the second step is an addition reaction to obtain intermediate $\mathbf{2}$

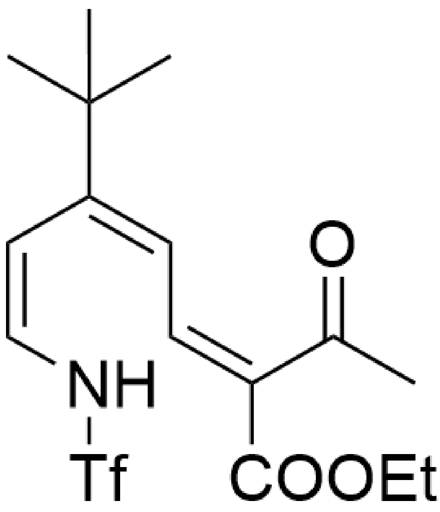


Intermediate **2**: CC(C)(C)C1=CC(C(C(OCC)=O)C(C)=O)N(S(=O)(C(F)(F)F)=O)C=C1

CHECKPOINT 1 PTS

Intermediate **2**: CC(C)(C)C1=CC(C(C(OCC)=O)C(C)=O)N(S(=O)(C(F)(F)F)=O)C=C1

Under the catalysis of excess base, further elimination reaction occurs to obtain intermediate **3**



Intermediate **3**: CC(C)(C)C(/C=C\NS(=O)(C(F)(F)F)=O)=C/C=C(C(OCC)=O)\C(C)=O

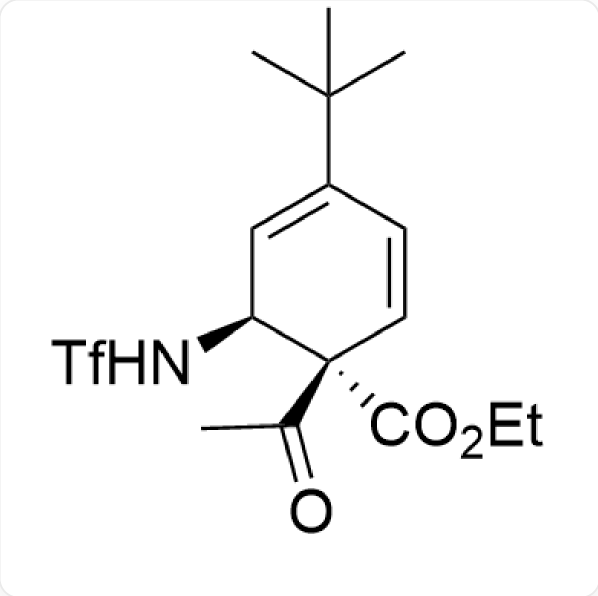
CHECKPOINT 1 PTS

Intermediate **3**: CC(C)(C)C(/C=C\NS(=O)(C(F)(F)F)=O)=C/C=C(C(OCC)=O)\C(C)=O (The cis/trans configuration of the double bond is not important)

Then, a six-electron electrocyclic reaction occurs to generate a pair of diastereomers. At this time, the ratio of products is approximately 1:1, but the molecular formulas of the two are still the same.

CHECKPOINT 1 PTS

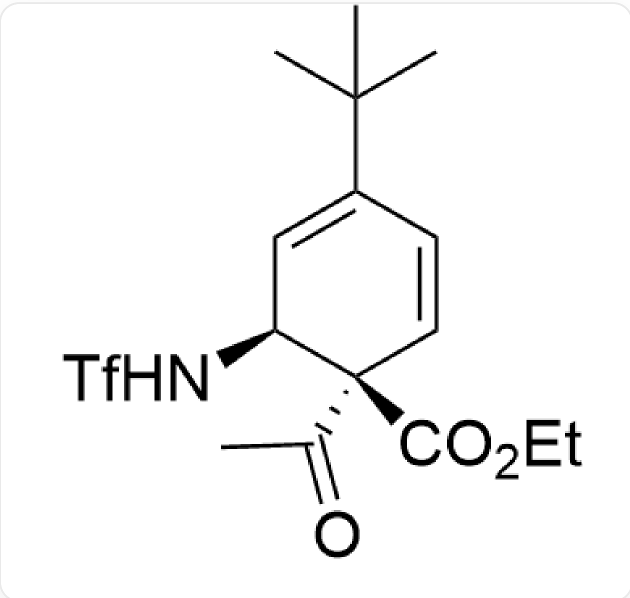
Then, a six-electron electrocyclic reaction occurs to generate a pair of diastereomers. At this time, the ratio of the obtained products is approximately 1:1, but the molecular formulas of the two are still the same.



Intermediate $\mathbf{4}$: CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@@](C(C)=O)(C(OCC)=O)C=C1

CHECKPOINT 1 PTS

Intermediate $\mathbf{4}$: CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@@](C(C)=O)(C(OCC)=O)C=C1



Intermediate $\mathbf{5}$: CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@](C(C)=O)(C(OCC)=O)C=C1

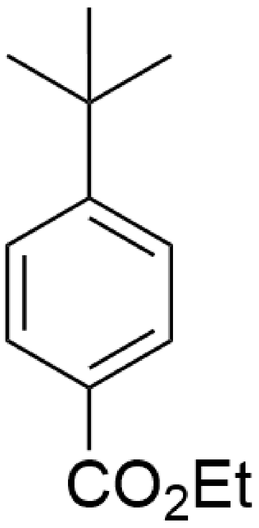
CHECKPOINT 1 PTS

Intermediate **5**: CC(C)(C)C1=C[C@H](NS(=O)(C(F)(F)F)=O)[C@](C(C)=O)(C(OCC)=O)C=C1

At this time, because the electrophilicity of the ketone is stronger than that of the ester, only intermediate **4** can further undergo a nucleophilic-elimination reaction to obtain product **B**

CHECKPOINT 1 PTS

At this time, because the electrophilicity of the ketone is stronger than that of the ester, only intermediate **4** can further undergo a nucleophilic-elimination reaction to obtain product **B**



Product **B**: CC(C)(C)C1=CC=C(C(OCC)=O)C=C1

CHECKPOINT 1 PTS

Product $\mathbf{B}$: CC(C)(C)C1=CC=C(C(OCC)=O)C=C1

Product $\mathbf{C}$ is intermediate $\mathbf{5}$